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Improve Accessibility and Visibility of Selected University Websites

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Abstract. Currently, there are millions of websites, but not all of them are accessible and visible in search engines, and in the different devices and technologies used. The purpose of this research is to 1) suggest an accessibility evaluation model according to the Web Content Accessibility Guidelines 2.1 standards; 2) correlate Webometrics and SCImago rankings to improve visibility, and 3) determine whether an accessible website can improve visibility. As a case study, the authors applied to 30 web pages from Latin American universities. The results indicate that although an educational institution is in the first rankings of the Webometrics and SCImago classification, they do not necessarily satisfy web accessibility requirements. This study can be a guide for the definition of institutional efforts to improve accessibility and visibility in the design of more inclusive and visible websites.

Keywords: Accessibility · University · Visibility · Websites · WCAG 2.1

1 Introduction

"We Are Social and Hootsuite" the digital reports [1] indicate that in the last 12 months internet users grew rapidly by 8.6%, these figures imply that there are 350 million new users and represent a total of 4,437 billion at the beginning of April 2019. As stated by the World Health Organization, there are more than one billion people, approximately 15% of the planet inhabitants [2], with some disability. Disability figures rise concerning people who age, which is why the number of chronic illnesses also rises.

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Currently, there are millions of websites, but not all are accessible, a website to achieve the popularity and success expected must be perceptible and accessible in different search engines, as well as in devices in such a way that supports assistive technologies. This study aims to assess the level of accessibility according to the standards of the Web Content Accessibility Guidelines (WCAG) 2.1 [3], and the perceptibility applied as a case study in 30 home pages of Latin American universities selected according to Webometrics[1] and SCImago[2] rankings.

According to SCImago, websites are essential and strategic for the dissemination of information and research carried out by an educational institution. Millions of users can visit a website more if it is more visible to search engines, which implies the need to increase the link with other websites. This study revealed that there are several problems related to levels of web accessibility, among which is the "Alt" attribute in images.

In this study we analyzed several parameters that help select Latin American universities, among the parameters investigated, are 1) the total volume of scientific production, 2) the relative quality, measured through impact indicators of the work, and 3) the percentage of documents that are published in journals classified by the level of impact in the first quartile of each category. Among the rankings, we have selected the SCImago ranking, which takes Scopus[3] as a principle, and the web ranking of Webometrics universities, which is used by Google Scholar.

The rest of the article is structured within the following way: in Sect. 2 we tend to depict the connected works, in Sect. 3 we tend to gift the methodology and also the case study, in Sect. 4 we tend to show the results and also the discussion, finally, in Sect. 5 we tend to embody the conclusions and future works derived from this analysis.

2 Background and Connected Work

Accessibility is the way users can communicate, interact, and navigate the web with ease. To better the level of accessibility, the Web Content Accessibility Guidelines 2.1 (WCAG 2.1) proposes 4 principles of accessibility, 13 guidelines and 78 compliance criteria, and some sufficient techniques and advisory techniques [3].

There are several studies related to web accessibility in higher education institutions, among the most recent there are: Ismail et al. [4] propose that in the analysis of accessibility of higher education websites it is essential to consider inclusion. The study applied the evaluation of sites with TAW. The authors noted 15995 warnings on the scale and 1356 unexamined items. Nagaraju et al. [5] analyzed 43 surveys on web accessibility that applied the WCAG 2.0. Initially, they identified similarities with previous and current research activities. They conducted a comparative study to determine accessibility barriers. Nir et al. [6] argued that because of the increasing range of pupils with incapacities in academic establishments, it is essential that policies

[1] http://www.webometrics.info/en/Latin_America.

[2] https://www.scimagoir.com/rankings.php?country=Latin%20America.

[3] https://www.scopus.com/.

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and variations square measure enforced to make sure accessibility to much range of users. The web pages examined presented accessibility barriers, the most frequent being "contrast," and "lack of alternative text."

In the authors' previous studies, Acosta-Vargas et al. [7] presented web accessibility problems identified in 348 Latin American university websites, consistent with the Webometrics ranking. In the assessment, the authors used the WAVE[4]; the results revealed that the websites of the universities included in this study violated the WCAG 2.0 guidelines. The authors propose strengthening accessibility policies in each country and applying the guidelines to make websites more inclusive.

The study [8] presents a collection of knowledge associated with the analysis of accessibility in 368 sites of selected institutions in Latin America taken from the Webometrics rankings. The dataset documents the foremost often errors in a method that alerts web developers, supporting them in making a lot of comprehensive and accessible websites for all kinds of individuals. The information shows that university websites have frequent issues associated with the lack of alternative text linked to images. The authors found that some of the websites violated WCAG 2.0 and 2.1 based accessibility requirements. The dataset was shared with public access for analysis and replication in the Mendeley repository[5].

Acosta-Vargas et al. [9] conducted studies for the analysis of web accessibility with heuristic ways for users with low vision. The study used a modification of the Barrier Walkthrough technique projected by Giorgio Brajnik, considering the WCAG 2.1. The modification was to incorporate persistence to work out the severity of the associate degree accessibility barrier. The tactic was applied to forty websites, together with those of thirty universities in Latin America, per the Webometrics ranking, and ten websites among the foremost visited, per the Alexa ranking. With this heuristic technique, the evaluators ended that though a website is in an exceedingly high ranking position, this does not imply that it is accessible and inclusive. However, the manual technique is just too long and, therefore, too expensive to unravel accessibility issues. This analysis might function as a start line for future studies associated with web accessibility heuristics.

SCimago Institutions Rankins [10] applies a methodology based on the classification of academic institutions related to research that in turn are classified by an indicator that combines three different sets of indicators based on 1) the performance of the research, 2) the results of the innovation and 3) the social impact measured by the visibility of its website [11]. SCImago considers several criteria [10], among them, 1) 50% of research, 2) 30% of innovation, and 3) 20% of the relationship with society. This methodology proposes 1) to define and identify a list of research institutions, 2) to assign publications and citations to each institution when considering the institutional affiliation of each author. On the other hand, Webometrics is an initiative of the Cybermetrics Laboratory; it applies four related indicators, 1) impact and visibility (50%); 2) presence (5%); 3) openness and transparency (10%), and 4) excellence (35%).

[4] https://wave.webaim.org/.

[5] https://data.mendeley.com/datasets/t6bmb24c9w/1.

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3 Method and Case Study

The case study applies to 30 universities that are in the top places in Latin America in the ranking of SCImago and Webometrics. The method applied to assess accessibility and visibility consists of six phases, as shown in Fig. 1, section left.

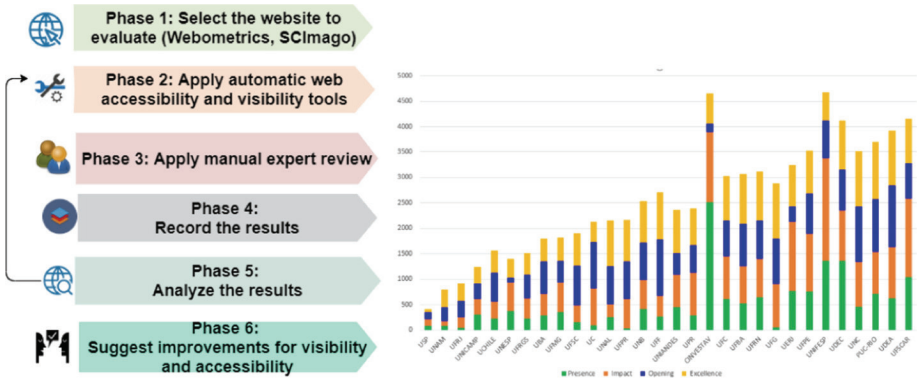


Fig. 1. Left: Method for assessing accessibility and visibility on a website. Right: Presence, impact, openness, and excellence components.

Table 1. Latin American universities, according to the Webometrics and SCImago.

LA Ranking	Acronym	Presence	Impact	Opening	Excellence	Errors
1	USP	80	127	137	70	36
2	UNAM	79	92	279	339	1
3	UFRJ	39	213	312	348	7
4	UNICAMP	292	319	298	331	2
5	UCHILE	225	330	570	428	1
6	UNESP	366	559	103	369	55
7	UFRGS	223	399	471	421	7
8	UBA	284	426	634	445	14
9	UFMG	346	584	435	458	6
10	UFSC	154	324	782	644	33
11	UC	91	723	913	406	3
12	UNAL	252	250	748	907	18
13	UFPR	30	574	738	817	22
14	UNB	414	563	740	821	26
15	UFF	259	407	1118	930	17
16	UNIANDES	441	648	423	854	45
17	UPR	286	845	544	714	0
18	CINVESTAV	2511	1385	165	593	75
19	UFC	603	850	702	875	3
20	UFBA	525	727	837	976	35

(*continued*)

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Table 1. (*continued*)

LA Ranking	Acronym	Presence	Impact	Opening	Excellence	Errors
21	UFRN	645	754	751	963	29
22	UFG	52	849	889	1092	16
23	UERJ	765	1359	304	813	34
24	UFPE	758	1128	794	844	14
25	UNIFESP	1360	2017	746	556	23
26	UDEC	1355	991	813	955	22
27	UNC	457	873	1099	1080	67
28	PUC-RIO	704	827	1035	1130	99
29	UDEA	618	1011	1218	1069	27
30	UFSCAR	1036	1550	690	873	22

Phase 1: Select the website to evaluate (Webometrics, SCImago); we select the Webometrics and SCImago websites of the Latin American universities that occupy the first places within the ranking. Table 1 shows the selected universities in Latin America. It contains the Latin American ranking, the acronym, presence, impact, opening, excellence, and Errors. *Phase 2: Apply automatic web accessibility and visibility tools*, in this phase, most pages of the websites of the chosen universities, extracted from Webometrics and SCImago, were evaluated, using the WAVE tool that identifies accessibility barriers. *Phase 3: Apply for manual expert review*; in this phase, two experts in web accessibility intervened to review the main page of each site, the data was recorded in the spreadsheet. *Phase 4: Record the results*, the data were recorded in a spreadsheet, all data from this case study was recorded in a spreadsheet, to share and replicate the experiment, the data set rests in a Mendeley repository. *Phase 5: Analyze the results*; in this part, the results within the Mendeley dataset were analyzed. Figure 2 shows that the universities that are in the highest-ranking do not necessarily have high scores in the components of presence, impact, openness and excellence; further details are provided in the results section. *Phase 6: Suggest improvements for visibility and accessibility*, to achieve an accessible and inclusive website we suggest 1) place a title in the head of the document; 2) encourage the exchange of content with buttons on social networks; 3) check the appearance of the website with a mobile and a tablet; 4) add an exclusive mobile style through media inquiries; 5) ensure the pages of the website embrace a meta description. The meta description can control how the search engine result will appear. 6) define significant headings for all web pages; the headlines help the placement of the website in the search engines and that visitors can browse the content. 7) enable GZIP compression on the webserver or in the application.

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4 Results and Discussion

Figure 1, section right shows that even though an institution is in the top rankings, it does not meet high levels of presence, impact, openness, and excellence. Figure 2, section left shows the University of Sao Paulo (USP) in the first place of the Webometrics ranking containing 36 errors of the websites when relating to Webometrics and SCImago. In the second place, the Universidad Nacional Autónoma de México (UNAM), presenting only one error. In third place is the Universidade Estadual Paulista Julio de Mesquita Filho (UNESP) that contains 55 errors despite being among the first three places in the ranking. Figure 2, section right shows the 30 universities evaluated with WAVE, where 29% of errors are related to the principle of understandable, in second place is 27% for operable, 25% for perceptible, and 19% for robust.

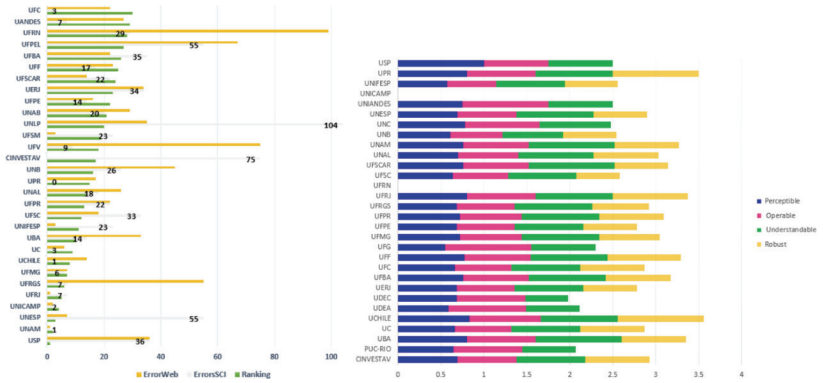


Fig. 2. Left: Accessibility errors when linking Webometrics and SCImago sites. Right: Principles of Web Accessibility

5 Conclusions and Future Works

This study revealed that there are several problems related to levels of web accessibility, among which is the "Alt" attribute, which helps search engines to recognize the images that web designers incorporate into web pages and help visually impaired users to use a screen reader that interprets the content or meaning of an image. Although technology has evolved a lot, search engines still have problems interpreting the content of images included in a website. Therefore, the attribute "Alt" is a very valuable parameter at the level of visibility and positioning; it helps us to generate large amounts of traffic on a website because the text will allow the images to be displayed in the results of Google Images[6]. As future work, we suggest looking at new methods of evaluating the accessibility of websites. Implement and disseminate WCAG 2.1 in

[6] https://www.google.com/imghp?hl=en.

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institutional policies for the design of more accessible websites. We recommend applying the suggestions made in this study to achieve higher visibility in search engines since, in future work, the method can be applied to achieve greater accessibility and visibility in government portals [12].

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